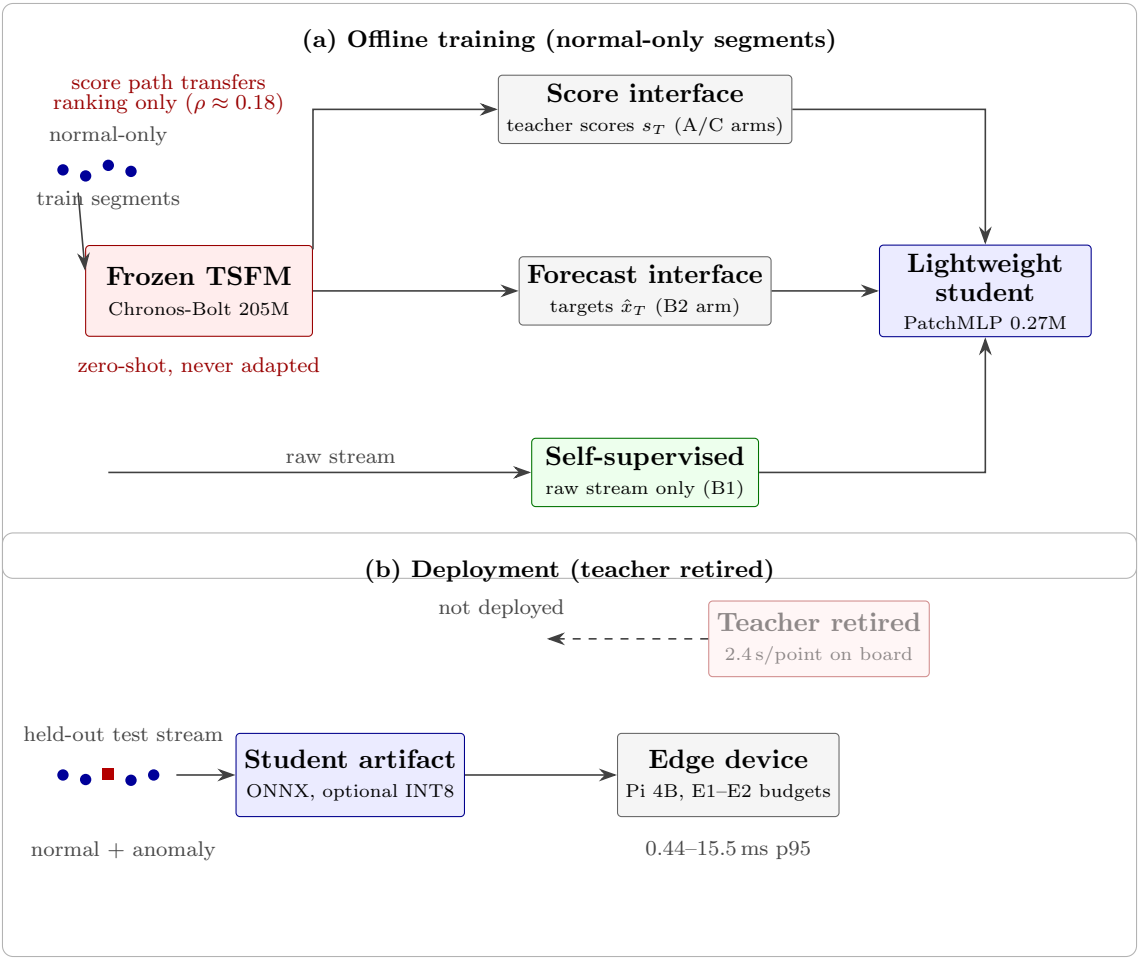

(a) Offline training (normal-only segments)
score path transfers
ranking only ($\rho \approx 0.18$)
normal-only
train segments
Frozen TSFM
Chronos-Bolt 205M
zero-shot, never adapted
Score interface
teacher scores s_T (A/C arms)
Forecast interface
targets $\hat{x}_T$ (B2 arm)
raw stream
Self-supervised
raw stream only (B1)
Lightweight student
PatchMLP 0.27M
(b) Deployment (teacher retired)
not deployed
Teacher retired
2.4 s/point on board
held-out test stream
normal + anomaly
Student artifact
ONNX, optional INT8
Edge device
Pi 4B, E1–E2 budgets
0.44–15.5 ms p95